

Laws of User Interface – Case Study Worksheet

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1. Jakob's Law

2. Fitts's Law

Scenario: A payment button in a banking app is very small and located far from the main thumb area.

3. Q: How does this violate Fitts's Law?

The small size and distant location of the payment button violate Fitts's Law, which states that the time to acquire a target increases with the distance to the target and decreases with the size of the target. Therefore, a small, distant button is slower to acquire.

4. Q: Suggest one redesign idea.

The payment button should be made larger and placed closer to the main thumb area to reduce movement time and increase the target's size, making it faster to acquire.

3. Hick's Law

Scenario: A streaming app offers 50+ categories on the home screen.

5. Q: Why might users feel overwhelmed?

Users may feel overwhelmed because Hick's Law states that the time it takes to make a decision increases with the number and complexity of choices. Offering 50+ categories on the home screen presents too many options, increasing the cognitive load.

6. Q: How can we simplify the interface?

The interface can be simplified by minimizing the choices shown initially, such as by highlighting only recommended or popular options, or by breaking the 50+ categories into smaller, logical groups or steps.

4. Law of Prägnanz

Scenario: An NGO website has overlapping shapes and bright animations. Users can't find the donation button.

7. Q: What visual issue occurs?

The overlapping shapes and flashy animations make the screen feel cluttered, making it hard for users to spot the donation button, which should stand out clearly against the background.

8. Q: Suggest a design improvement.

Remove the overlapping shapes and bright animations to reduce visual noise. Ensure the donation button is a simple, distinct shape with high contrast against its background, allowing users to perceive it immediately as the primary figure.

5. Law of Common Region

Scenario: Assignment details and "Submit" buttons are placed far apart in a course system.

9. Q: What relationship problem does this cause?

When the assignment details are far from the "Submit" button, users might not realize they belong together, making it unclear which task the button will submit.

10. Q: How can grouping improve usability?

Grouping the assignment details and the "Submit" button within a defined, shared boundary will visually link them, clearly indicating that the button is specifically for submitting the associated assignment.

6. Law of Proximity

Scenario: Flight price and "Book" buttons are separated by a lot of space.

11. Q: What confusion might this cause?

Placing the flight price far from the "Book" button can confuse users, as they expect related information to be close making it harder to link the button to the right flight.

12. Q: How can proximity fix it?

Placing the flight price and the "Book" button immediately next to each other (or closer than to other elements) will use the Law of Proximity to visually group them, making their relationship instantly clear and reducing scanning time.

7. Law of Similarity

Scenario: All buttons (Save, Delete, Update) look identical.

13. Q: How might this confuse users?

When all buttons look the same, users can easily mix them up making them more likely to click "Delete" when they meant "Save" because similar-looking elements are seen as having the same function.

14. Q: Suggest a differentiation method.

Differentiate the buttons based on their function, for example, by using different colors (eg green for "Save/Update," red for "Delete") or distinct icons to reflect their unique actions, thereby overriding the Law of Similarity for critical functions.

8.Serial Position Effect

Scenario: A 10-step tutorial hides the most important information in the middle.

15. Q: How can step order affect memory?

The step order affects memory because the Serial Position Effect predicts that users are most likely to remember the first and last items in a series. Hiding important information in the middle means it is less likely to be recalled.

16. Q: How would you reorder it?

The most important information should be moved to the beginning of the tutorial (primacy) and/or the end (recency) to capitalize on the Serial Position Effect and maximize the chances of the user remembering it.

9.Aesthetic–Usability Effect

Scenario: A finance app works perfectly but looks outdated.

17. Q: Why do users still dislike it?

Users dislike the app because, according to the Aesthetic–Usability Effect, looks matter even a perfectly working app feels less trustworthy and enjoyable if its design looks old or outdated.

18. Q: What design updates can change perception?

Use a modern, clean design with fresh colors, clear fonts, and quality visuals to instantly make the app look more professional, trustworthy, and pleasant to use.

10.Doherty Threshold

Scenario: A chatbot takes 4 seconds to reply with no typing indicator.

19. Q: How does this affect engagement?

A 4-second delay with no feedback breaks the user's flow and makes the system feel unresponsive. The Doherty Threshold shows that keeping interactions quick and responsive keeps users focused and satisfied.

20. Q: Suggest a UI fix.

Provide immediate UI feedback for the user's action within the 400-millisecond threshold, such as a typing indicator or a simple loading spinner (progress indicator) to let the user know the system has received the request and is processing the reply.

11. Miller's Law

Scenario: A sign-up form has 15 fields in one screen.

21. Q: Why do users quit midway?

Users drop off because the long 15-field form overloads their memory and focus. According to Miller's Law, people can only handle a few pieces of information at once, so such a lengthy form feels overwhelming and discouraging.

22. Q: Suggest a better form structure.

Break the 15 fields into smaller, logical chunks (e.g., "Personal Info," "Account Details," "Address") across multiple screens or steps. This structured, progressive approach minimizes cognitive load and makes the form seem less daunting.

12. Tesler's Law

Scenario: A hospital system forces nurses to manually validate all data entries.

23. Q: What complexity issue is present?

The system violates Tesler's Law by pushing too much complexity onto nurses, making them handle manual data checks that increase workload and the risk of errors instead of simplifying the process through automation.

24. Q: How can automation help?

Automating routine tasks like data validation, error checking, and data formatting helps the system absorb the complexity itself, reducing the burden on the nurses and allowing them to focus on critical tasks, thus making the overall experience simpler for the user.

13. Von Restorff Effect (Isolation Effect)

Scenario: All product cards look identical; users ignore "Sale" items.

25. Q: How can isolation increase visibility?

The Von Restorff Effect (or isolation effect) predicts that the single object that differs from multiple similar objects is the most likely to be remembered. Isolation can increase the visibility of "Sale" items by making them visually distinct from all other product cards.

26. Q: What visual cue would you add?

Add a strong visual cue that isolates the "Sale" product card, such as a contrasting color background, a prominent "SALE" badge, or a distinct larger size to make it stand out from the similar, non-sale product cards.

14. Reflection

Which UI law do you find most useful as a designer, and why?

I find Fitts's Law most useful because it gives a clear, practical way to design faster and easier interactions. By simply making important buttons larger and closer, users can navigate more quickly and accurately, improving both speed and overall usability.

Scenario-II

EduLearn is an online course platform for college students. Its dashboard screen has the following issues:

- The “**Start Lesson**” button is **small and placed at the top-left** of the screen.
- There are **15 menu options** listed in one vertical column without grouping.
- The **progress bar** and **course completion percentage** are located **at the very bottom**, often missed by users.
- The screen uses **five different colors and fonts**, making it visually busy.
- Important options like “**Continue Course**” and “**Join Live Session**” look the same as lesser-used ones like “**View Policy.**”
- When users click “**Start Lesson,**” the screen takes **4–5 seconds to load** with no feedback.

1. List all the usability or design problems visible in the above screen.

- Inefficiency: The "Start Lesson" button is small and located far from the user's main interaction area (top-left).

- Cognitive Overload: Listing 15 menu options in a single column without grouping overwhelms the user.

- Poor Visibility: The progress bar and completion percentage, which are key feedback elements, are located at the very bottom and often missed.

- Visual Clutter: The use of five different colours and fonts makes the screen look busy and unprofessional.

- Lack of Prioritization: Important options ("Continue Course," "Join Live Session") look identical to minor options ("View Policy").

- Slow/Uncertain Feedback: A 4–5 second load time after clicking a button with no visual feedback leaves the user waiting and uncertain.

2. For each problem, identify which UI/UX law(s) are being violated.

- Small and distant primary button - Fitts's Law (Time to acquire a target is a function of distance and size).

- 15 menu options in one column - Miller's Law (Too many items exceed working memory capacity) / Hick's Law (Decision time increases with the number of choices).

- Progress bar at the very bottom - Serial Position Effect (Users best remember the first and last items in a series).

- Five colors and fonts (visually busy) - Law of Prägnanz (The tendency to perceive the simplest, most stable figure).

- Important and lesser-used options look the same - Von Restorff Effect (The item that differs is most likely to be remembered) / Law of Similarity.

- 4–5 seconds load time with no feedback - Doherty Threshold (Productivity suffers when waiting time exceeds 400ms without feedback).
3. Students propose concrete design actions guided by these laws.
 - Fitts's Law - Increase the size of the "Start Lesson" button and move it to the bottom-center area of the screen.
 - Miller's/Hick's Law - Group the 15 menu options into smaller, logical chunks (4-5 categories) to minimize cognitive load.
 - Serial Position Effect - Relocate the progress bar and completion percentage to the top of the screen (the beginning of the series) to ensure it is immediately seen and remembered.
 - Law of Prägnanz - Simplify the visual design by reducing the palette to one primary colour and using one or two complementary fonts for clarity.
 - Von Restorff Effect - Apply the Isolation Effect: make "Continue Course" and "Join Live Session" visually distinct with a high-contrast color and a button style that differs from "View Policy".
 - Doherty Threshold - Provide immediate feedback (within 400ms) upon click, such as a loading spinner or progress indicator, to bridge the 4–5 second waiting period.
 4. Which three laws had the most impact on the redesign?
 - Fitts's Law: Fixing the small, distant primary action button is crucial for basic efficiency and speed, directly improving the user's core task flow (starting a lesson).
 - Hick's Law (and Miller's Law): Addressing the overwhelming 15-option menu is essential for reducing cognitive load and preventing the user from abandoning the app due to choice paralysis.
 - Von Restorff Effect: This law ensures that the most important calls-to-action ("Continue Course") are prioritized and visible, effectively guiding the student's attention to their main task.
 5. How did combining multiple UX laws result in a better user experience?

By applying Fitts's and Doherty's Laws, the dashboard became faster and more responsive. Hick's and Prägnanz's Laws made it cleaner and easier to understand, while the Von Restorff and Serial Position Effects helped users notice and remember key actions effortlessly.
 6. What trade-offs might arise when applying too many principles at once?

Applying too many design laws at once can backfire like oversized buttons may clutter the screen, too many "highlighted" elements can create visual chaos, and oversimplifying menus can make navigation slower instead of easier.